

Questionnaire results and interview summaries per participant

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Participant 001

Statement	User-initiated help	System-initiated help
The help supported me during the task.	4	5
I found the way help was offered pleasant.	4	3
I would want to use this type of help again for a similar task.	5	4
The help appeared at unwanted moments.	2	4
How mentally demanding was the task?	3	3
How much effort did you have to put in to complete the task?	2	4
How frustrated did you feel during the task?	3	3

Table 1: Questionnaire scores for participant 001.

Interview summary:

- The participant considered both assistance methods easy to understand and appreciated the simple language used throughout the task.
- The participant viewed both assistance methods positively and believed that a system like this could help older adults when using computers.
- During the user-initiated condition, the participant found the assistance helpful because it indicated mistakes and provided an opportunity to correct the sequence.
- During the system-initiated condition, the participant felt that assistance was offered too quickly and suggested that older adults may need more time to think before receiving support.
- The participant appreciated that automatic assistance responded when mistakes were made and reduced the need for guessing.
- The participant valued having control over when assistance was provided and preferred being able to decide independently when help was needed.
- Overall, the participant preferred the user-initiated condition and indicated that in everyday situations they would only ask for help when unable to solve a problem independently.

Participant 002

Statement	User-initiated help	System-initiated help
The help supported me during the task.	4	4
I found the way help was offered pleasant.	4	4
I would want to use this type of help again for a similar task.	4	4
The help appeared at unwanted moments.	2	2
How mentally demanding was the task?	1	3
How much effort did you have to put in to complete the task?	3	2
How frustrated did you feel during the task?	2	2

Table 2: Questionnaire scores for participant 002.

Interview summary:

- The participant responded positively to both assistance methods and described the interaction as friendly and enjoyable.
- The participant appreciated that mistakes could be corrected without negative consequences and felt that this reduced pressure and stress during the task.
- During the system-initiated condition, the participant found the assistance pleasant and appreciated receiving support when making mistakes.
- The participant noted that some hints were not sufficiently clear and suggested that more specific guidance could improve the assistance.
- During the user-initiated condition, the participant appreciated being able to click on the assistant when needed and found the feedback helpful for correcting mistakes.
- The participant valued having control over when assistance was provided and preferred being able to decide independently whether help was needed.
- The participant described the system-initiated condition as somewhat more present and less calm than the user-initiated condition.
- Overall, the participant preferred the user-initiated condition because it felt more independent and less intrusive.
- The participant indicated that system-initiated assistance could still be useful in unfamiliar situations where additional guidance may be needed.

Participant 003

Statement	User-initiated help	System-initiated help
The help supported me during the task.	5	4
I found the way help was offered pleasant.	4	3
I would want to use this type of help again for a similar task.	4	4
The help appeared at unwanted moments.	3	2
How mentally demanding was the task?	4	3
How much effort did you have to put in to complete the task?	4	4
How frustrated did you feel during the task?	5	4

Table 3: Questionnaire scores for participant 003.

Interview summary:

- The participant found the task mentally demanding and reported putting considerable effort into completing it.
- During the user-initiated condition, the participant considered the assistance clear and helpful because it provided useful tips when needed.
- The participant indicated that negative feedback about mistakes was sometimes difficult to receive, although this was mainly related to making errors rather than to the assistance itself.
- During the system-initiated condition, the participant felt that some feedback was unnecessary because they were already aware that a mistake had been made.
- The participant considered the system-initiated assistance helpful but felt that it was sometimes offered too early.
- The participant reported that both assistance methods supported task completion and helped by providing guidance during the task.
- The participant preferred the user-initiated condition because it allowed additional time for independent problem solving before assistance was provided.

Participant 004

Statement	User-initiated help	System-initiated help
The help supported me during the task.	4	4
I found the way help was offered pleasant.	4	4
I would want to use this type of help again for a similar task.	4	3
The help appeared at unwanted moments.	1	1
How mentally demanding was the task?	1	1
How much effort did you have to put in to complete the task?	2	2
How frustrated did you feel during the task?	1	1

Table 4: Questionnaire scores for participant 004.

Interview summary:

- The participant responded positively to both assistance methods and considered both approaches useful.
- During the system-initiated condition, the participant felt that assistance was sometimes offered too quickly and would have preferred more time to think independently before receiving support.
- The participant appreciated the gradual structure of the assistance and found the different hint levels helpful.
- The participant found the hints useful because they provided guidance when mistakes were made.
- During the user-initiated condition, the participant initially had to adjust to requesting help independently but quickly became comfortable with this approach.
- The participant valued having the freedom to decide whether assistance was needed and appreciated being able to choose when help was provided.
- Overall, the participant preferred the user-initiated condition because it offered greater control while still providing support when needed.

Participant 005

Statement	User-initiated help	System-initiated help
The help supported me during the task.	5	5
I found the way help was offered pleasant.	5	4
I would want to use this type of help again for a similar task.	5	3
The help appeared at unwanted moments.	1	1
How mentally demanding was the task?	2	2
How much effort did you have to put in to complete the task?	3	3
How frustrated did you feel during the task?	3	1

Table 5: Questionnaire scores for participant 005.

Interview summary:

- The participant responded positively to both assistance methods and reported few difficulties during the task.
- During the user-initiated condition, the participant appreciated being able to review and adjust answers after receiving feedback.
- The participant found the assistance helpful because it indicated when mistakes had been made and provided an opportunity to correct them.
- During the system-initiated condition, the participant appreciated that assistance was offered at appropriate moments and helped identify errors during the task.
- The participant liked that the system offered help while still allowing a choice between accepting assistance or continuing independently.
- The participant considered both assistance methods useful and pleasant to use.
- Overall, the participant preferred the system-initiated condition because it was easier to use and removed the need to actively ask for help.

Participant 006

Statement	User-initiated help	System-initiated help
The help supported me during the task.	4	4
I found the way help was offered pleasant.	4	4
I would want to use this type of help again for a similar task.	4	4
The help appeared at unwanted moments.	3	2
How mentally demanding was the task?	3	3
How much effort did you have to put in to complete the task?	3	3
How frustrated did you feel during the task?	4	4

Table 6: Questionnaire scores for participant 006.

Interview summary:

- The participant reported that both assistance methods were helpful during the task.
- The participant considered the assistance useful because it helped guide attention towards the correct solution.
- During the system-initiated condition, the participant appreciated being redirected when difficulties occurred.
- The participant reported experiencing some frustration during the task but attributed this primarily to personal performance rather than the assistance itself.
- No specific negative aspects of either assistance method were reported.
- Overall, the participant preferred the user-initiated condition because it allowed the task to be completed more independently and without unsolicited support.
- The participant indicated a general preference for solving problems independently rather than receiving automatic assistance.

Participant 007

Statement	User-initiated help	System-initiated help
The help supported me during the task.	5	4
I found the way help was offered pleasant.	5	4
I would want to use this type of help again for a similar task.	5	3
The help appeared at unwanted moments.	1	1
How mentally demanding was the task?	3	3
How much effort did you have to put in to complete the task?	3	3
How frustrated did you feel during the task?	1	1

Table 7: Questionnaire scores for participant 007.

Interview summary:

- The participant responded positively to both assistance methods and described the assistance as pleasant and easy to understand.
- During the user-initiated condition, the participant appreciated that the feedback was clear and easy to interpret.
- During the system-initiated condition, the participant valued having the opportunity to first attempt the task independently before deciding whether to accept assistance.
- The participant reported that the user-initiated condition felt more like direct assistance, whereas the system-initiated condition felt more open and allowed greater freedom in deciding whether help was needed.
- The participant indicated that both assistance methods were helpful and that the assistance supported task completion.
- No negative aspects of either assistance method were reported.
- Although the participant felt that the user-initiated condition provided more direct support, the participant ultimately preferred the system-initiated condition.

Participant 008

Statement	User-initiated help	System-initiated help
The help supported me during the task.	4	4
I found the way help was offered pleasant.	4	4
I would want to use this type of help again for a similar task.	4	4
The help appeared at unwanted moments.	4	4
How mentally demanding was the task?	4	3
How much effort did you have to put in to complete the task?	4	1
How frustrated did you feel during the task?	4	3

Table 8: Questionnaire scores for participant 008.

Interview summary:

- The participant responded positively to both assistance methods and considered both approaches helpful.
- During the system-initiated condition, the participant appreciated being informed immediately when a mistake had been made.
- The participant described the automatic assistance as convenient and found it helpful that errors were identified without requiring additional action.
- During the user-initiated condition, the participant indicated that assistance was less useful when the sequence had already been forgotten.
- The participant reported that both assistance methods supported task completion.
- No negative aspects of the assistance itself were reported.
- Overall, the participant preferred the system-initiated condition because it provided immediate feedback and allowed mistakes to be corrected more easily.

Participant 009

Statement	User-initiated help	System-initiated help
The help supported me during the task.	4	5
I found the way help was offered pleasant.	4	4
I would want to use this type of help again for a similar task.	5	4
The help appeared at unwanted moments.	1	1
How mentally demanding was the task?	3	3
How much effort did you have to put in to complete the task?	4	4
How frustrated did you feel during the task?	3	3

Table 9: Questionnaire scores for participant 009.

Interview summary:

- The participant responded positively to both assistance methods and considered both approaches useful.
- During the user-initiated condition, the participant appreciated being able to review information and request assistance when needed.
- The participant found the user-initiated assistance helpful because it provided an opportunity to continue the task after becoming uncertain.
- During the system-initiated condition, the participant described the assistance as easy to use and appreciated being informed quickly when mistakes occurred.
- The participant valued receiving immediate feedback because it made it easier to identify and correct errors.
- No negative aspects of either assistance method were reported.
- Overall, the participant preferred the system-initiated condition because feedback was provided more quickly and efficiently.

Participant 010

Statement	User-initiated help	System-initiated help
The help supported me during the task.	5	3
I found the way help was offered pleasant.	5	4
I would want to use this type of help again for a similar task.	5	4
The help appeared at unwanted moments.	1	1
How mentally demanding was the task?	1	4
How much effort did you have to put in to complete the task?	2	4
How frustrated did you feel during the task?	1	2

Table 10: Questionnaire scores for participant 010.

Interview summary:

- The participant reported that assistance was necessary to successfully complete the task.
- During the system-initiated condition, the participant considered the assistance helpful because it indicated when mistakes had been made.
- The participant felt that the system-initiated assistance only provided partial support when the sequence had been completely forgotten.
- During the user-initiated condition, the participant found the assistance much more helpful and appreciated being able to request support whenever needed.
- The participant reported feeling more confident during the user-initiated condition because help was always available when uncertainty arose.
- The participant valued being able to repeatedly check and verify answers before completing the task.
- No negative aspects of either assistance method were reported.
- Overall, the participant preferred the user-initiated condition because it provided greater control and made it easier to obtain help when needed.

Participant 011

Statement	User-initiated help	System-initiated help
The help supported me during the task.	3	4
I found the way help was offered pleasant.	4	4
I would want to use this type of help again for a similar task.	4	5
The help appeared at unwanted moments.	1	2
How mentally demanding was the task?	4	2
How much effort did you have to put in to complete the task?	5	3
How frustrated did you feel during the task?	4	2

Table 11: Questionnaire scores for participant 011.

Interview summary:

- The participant considered both assistance methods useful and appreciated having support available during the task.
- During the user-initiated condition, the participant found some hints unclear, particularly references to the upper and lower row.
- The participant reported feeling distracted by the avatar during the user-initiated condition.
- During the system-initiated condition, the participant found the assistance much more pleasant and easier to use.
- The participant appreciated that the system responded more quickly and provided more direct feedback.
- The participant reported feeling less distracted during the system-initiated condition because the assistance was more to the point.
- Overall, the participant preferred the system-initiated condition because it was faster, clearer, and less distracting.

Participant 012

Statement	User-initiated help	System-initiated help
The help supported me during the task.	4	4
I found the way help was offered pleasant.	5	4
I would want to use this type of help again for a similar task.	4	4
The help appeared at unwanted moments.	2	2
How mentally demanding was the task?	3	2
How much effort did you have to put in to complete the task?	4	4
How frustrated did you feel during the task?	4	4

Table 12: Questionnaire scores for participant 012.

Interview summary:

- The participant responded positively to both assistance methods and considered both approaches helpful.
- During the system-initiated condition, the participant appreciated receiving hints and found the assistance supportive.
- The participant reported that the hints helped guide task performance when difficulties occurred.
- During the user-initiated condition, the participant found the assistance more convenient because help could be requested whenever desired.
- The participant particularly appreciated being able to repeatedly check progress and verify performance during the task.
- The participant reported no negative aspects of either assistance method. Difficulties were mainly related to completing the memory task itself rather than to the assistance.
- Overall, the participant preferred the user-initiated condition because it felt more convenient and allowed greater control over when assistance was used.